



UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II, SESSION 2024/2025

PAPER 2 (PRACTICAL)

SUBJECT CODE : SECJ1023
SUBJECT NAME : PROGRAMMING TECHNIQUE II
SECTION : 01 / 02 / 03 / 04 / 05 / 06 / 07
TIME : (3 HOURS)
DATE/DAY :
VENUES :

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

INSTRUCTIONS:

1. This is a CLOSED-BOOK practical exam. You will attempt the exam by coding on a computer.
2. You will be provided with some reference resources. You are only allowed to refer to the provided resources.
3. References to other resources including any source code, any websites, etc., are strictly prohibited.
4. The use of any kind of artificial intelligence tool is strictly prohibited. These tools include ChatGPT, AI-based VS Code extensions such as Co-Pilot, etc.
5. You should code offline using a C++ IDE such as VS Code, Dev C++, etc.
6. You must use a PC provided in the lab. The use of your own computer is not allowed.
7. You will only use e-learning to download the provided resources. Please refer to the e-learning for your section.
8. **All COMMENTS in your program WILL NOT BE EVALUATED.**
9. Programs that **CANNOT COMPILE** will be **penalized by 50%**.
10. Be warned that your program will be subject to a screening process for **PLAGIARISM DETECTION**.
11. **NO TECHNICAL ASSISTANCE** will be provided during the exam. This also applies to problems such as the IDE not working, the program not compiling, etc. If you encounter a problem, you must solve it yourself.
12. All questions related to the exam question will not be entertained at all. Please read the question carefully.
13. This examination book consists of TWO (2) questions. **ANSWER ALL QUESTIONS.**
14. A starter code is provided only for Question 1. As for Question 2, you will have to write the program from scratch.

Name	
IC/ISID No.	
Year / Course	
Section	
Lecturer Name	

WARNING: Disciplinary action will be taken against students who are found cheating during the examination.

This question paper consists of seven (7) printed pages including this page.

QUESTION 1

[35 MARKS]

Instructions:

- A starter code is provided for this question in **program1.cpp**.
- For questions (a) to (f), all modifications should be made within the same file.
- Please adhere to the following guidelines:
 - Use inline coding style only.
 - Do not modify any access specifiers in the program.
 - Do not add new attributes or methods to any class.
 - Do not modify the headers of existing methods or functions in the program.

A class diagram modelling human resources management is provided in **Figure 1**. Answer all the following questions:

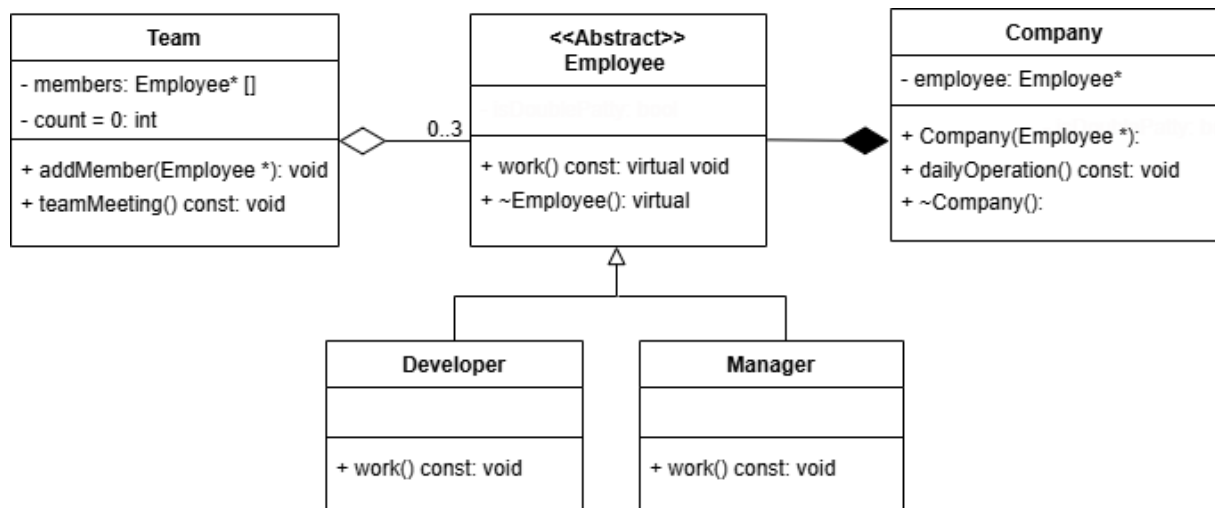


Figure 1

a) Refer to the class diagram and determine the Object-Oriented Programming relationships between the classes listed below. **Note:** If no relationship exists, write “No relationship”. Put your answers in the program code at Line 11, Line 14, Line 17, and Line 20, respectively.

- Developer and Employee
- Manager and Team
- Company and Employee
- Team and Employee

(4 marks)

- b) Complete the `Employee` class (as a base class) by implementing the following methods:
- `virtual void work() const`
 - Print "Employee working".
 - This function should be overridden by derived classes (`Developer` and `Manager`).
 - `virtual ~Employee()`
 - Virtual destructor for destroying derived class objects (`Developer` and `Manager`) correctly. (3 marks)
- c) Complete the `Company` class by implementing the following methods:
- `Company(Employee *)`
 - Constructor that accepts and owns a pointer to an `Employee`.
 - `~Company()`
 - Destructor that deletes the owned `Employee`.
 - `void dailyOperation() const`
 - Print "Company operation: ".
 - Then, calls the `work()` method of the employee. (7 marks)
- d) Complete the `Team` class by implementing the following methods:
- `void addMember(Employee*)`
 - Adds the given `Employee` pointer to the team.
 - Limit: Maximum 3 members only.
 - `void teamMeeting() const`
 - Iterates through all members and calls their respective `work()` methods. (8 marks)
- e) Modify the `main()` function to demonstrate polymorphism:
- Call the `work()` method for each `Employee` pointer.
 - Create a `Company` object and pass a dynamically allocated `Developer` object to it. Then, call `dailyOperation()` method.
 - Add both dynamically allocated `Developer` and `Manager` objects to the team using `addMember()` function. Then, call `teamMeeting()` function to demonstrate polymorphic behavior. (10 marks)

- f) Ensure your program produces exactly the following output, without any compilation or runtime errors:

Developer writing code

Manager in meeting

Company operation: Developer writing code

Developer writing code

Manager in meeting

(3 marks)

QUESTION 2

[65 MARKS]

Instructions:

- No starter code is provided for this question. You will have to write the program from scratch.
- Write a complete C++ program and save it in a single file named **program2.cpp**.

Write a complete C++ program based on the UML class diagram in **Figure 2**. Implement all the classes with their member variables (attributes) and member functions (methods) as shown in the diagram. The purpose of each function is as the name implies, and more explanations are provided below. Your program should be able to produce the output as shown in **Figure 3**.

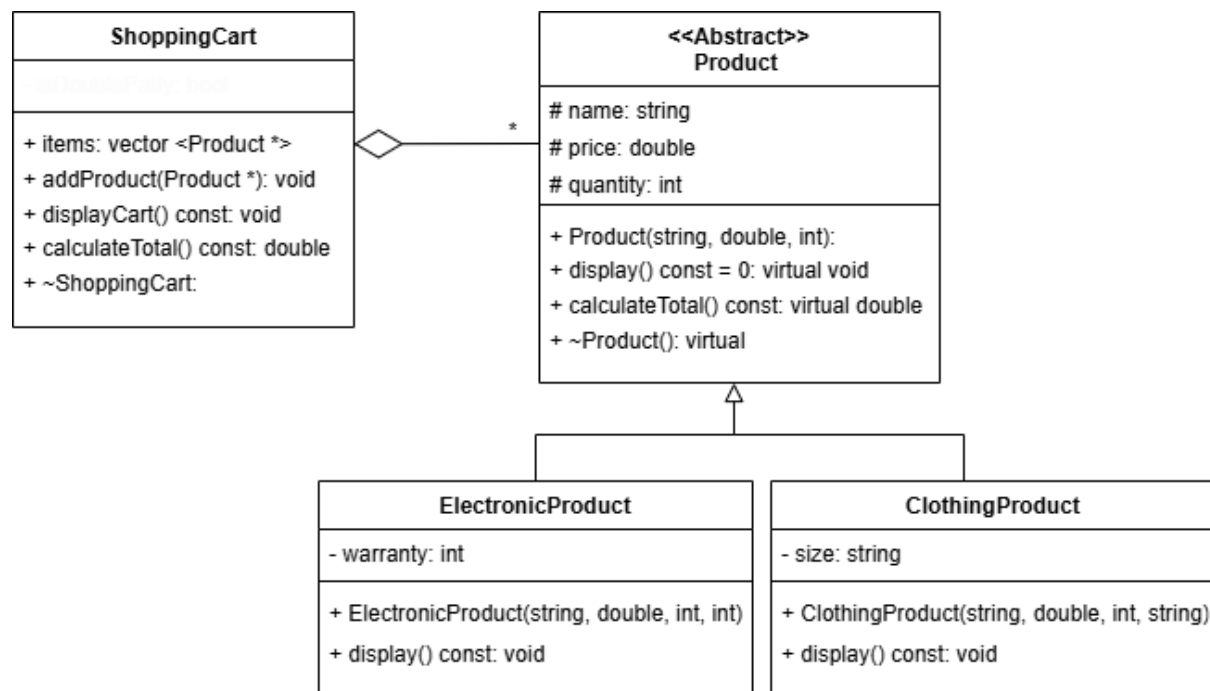


Figure 2: UML class diagram

- a) Define the abstract class `Product`. This class must include the following four member functions:
- A constructor with parameters to initialize the attributes `name`, `price`, and `quantity`.
 - A pure virtual function `display()` to be overridden by derived classes.
 - A virtual function `calculateTotal()` that returns the total value of the product based on its price and quantity.

- iv) A virtual destructor that does not perform any specific action. (11 marks)
- b) Define the `ElectronicProduct` class as a subclass of the `Product` class. This class should include the following two member functions:
- i) A constructor to initialize all member attributes including those inherited from the `Product` class.
 - ii) A `display()` function to show the product's name, price, quantity, and warranty as illustrated in **Figure 3**. (11 marks)
- c) Define the `ClothingProduct` class as a subclass of the `Product` class. This class should include the following two member functions:
- i) A constructor to initialize all member attributes including those inherited from the `Product` class.
 - ii) A `display()` function to show the product's name, price, quantity, and size as illustrated in **Figure 3**. (11 marks)
- d) Define the `ShoppingCart` class. This class should have five members:
- i) A vector stores the list of product items in the cart.
 - ii) A `addProduct()` function to add a product pointer to the cart.
 - iii) A `displayCart()` function to display all items in the cart as illustrated in **Figure 3**.
 - iv) A `calculateTotal()` function to return the total value of the cart by summing all product totals.
 - v) A destructor to release a memory for the `Product` objects stored in the vector. (14 marks)
- e) Define the `main()` function:
- i) Create a `ShoppingCart` object.
 - ii) Dynamically create at least one `ElectronicProduct` and one `ClothingProduct` objects. **Note:** Use the concept of polymorphism.
 - iii) Add all the created objects in (ii) into the cart.
 - iv) Display the details of all the products in the cart as depicted in **Figure 3**.
 - v) Calculate and display the total purchase amount as illustrated in **Figure 3**. (8 marks)

- f) Add appropriate statements to the program in (e) to handle an error if the total price is below RM100.00.
- i) Use exception handling (`try`, `throw`, `catch`) to detect and handle the error.
 - ii) If the total is less than RM100.00, throw an exception with a relevant error message.
 - iii) Catch the exception and display the message as shown in **Figure 4**. (5 marks)
- g) Add appropriate statements to the program in (e) to count the number of products in the cart with total price is greater than RM50.00. (5 marks)

```
Electronic: Smart Watch  
Price: RM120  
Quantity: 1  
Warranty: 24 months  
  
Clothing: T-Shirt  
Price: RM35  
Quantity: 2  
Size: L  
  
Total Amount: RM190  
Number of products above RM50: 2
```

Figure 3: Program output

```
LowTotalException: Your order must be at least RM100 to proceed.
```

Figure 4: Sample output of error handling

~ End of questions ~